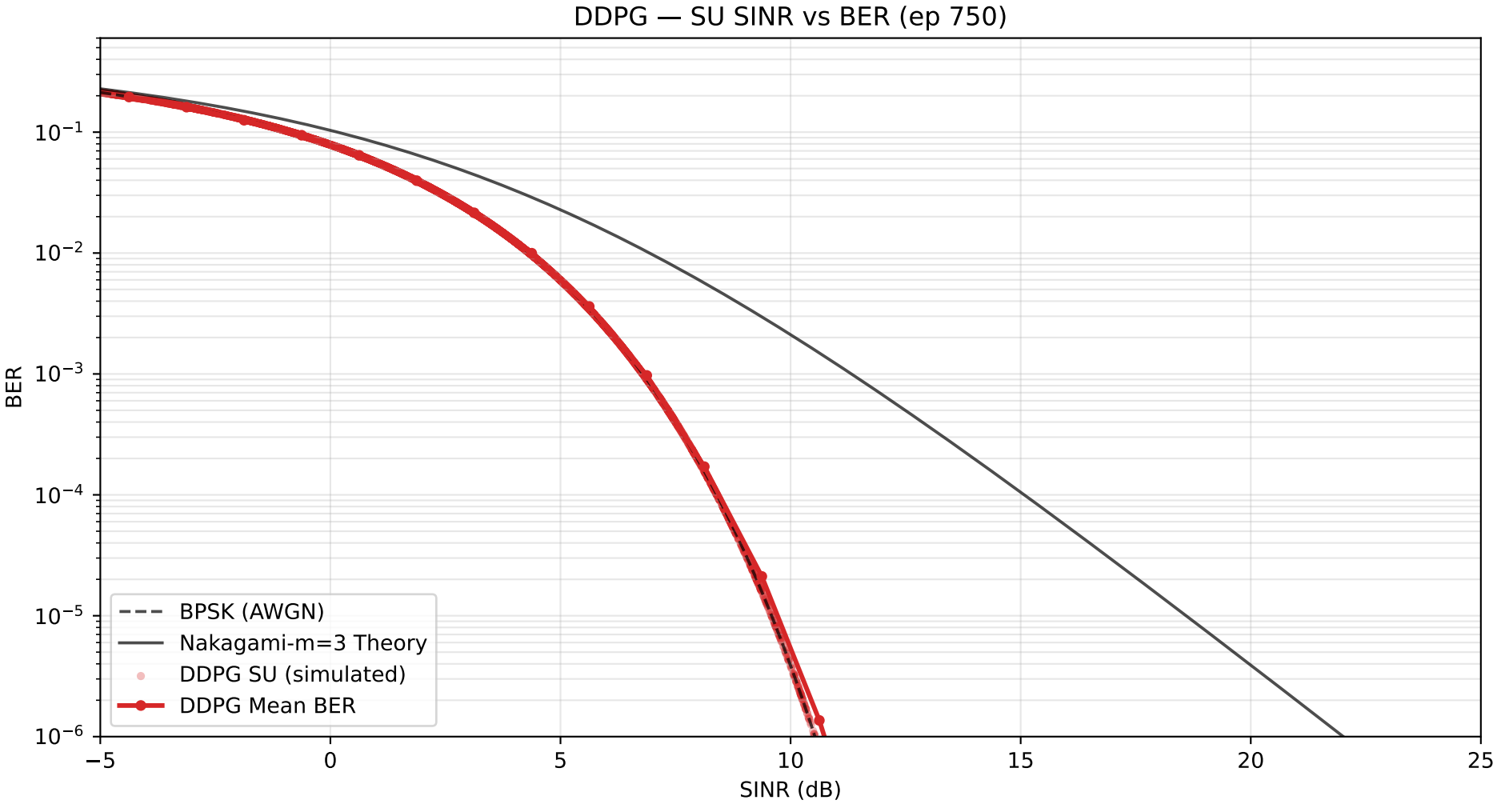
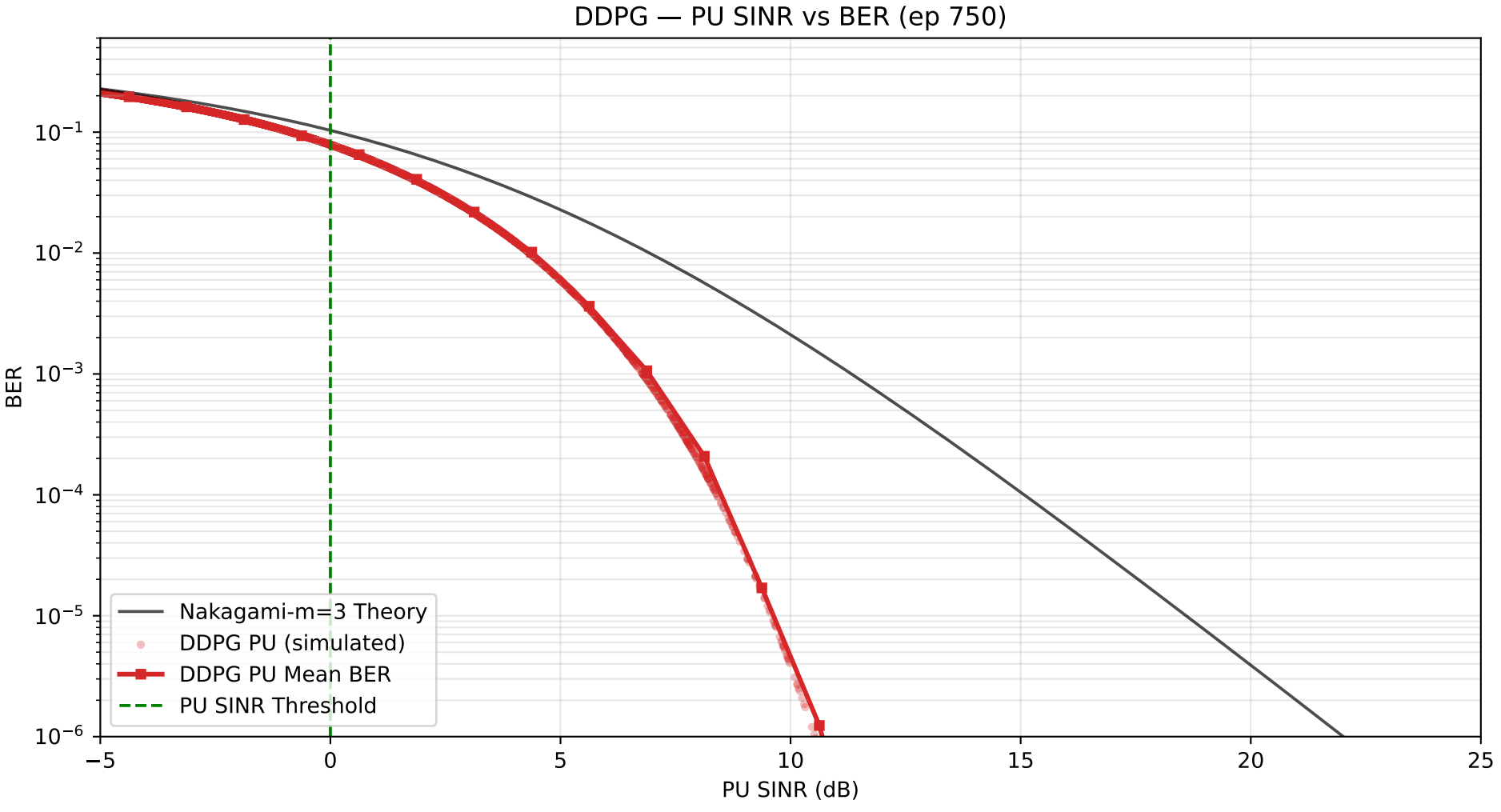
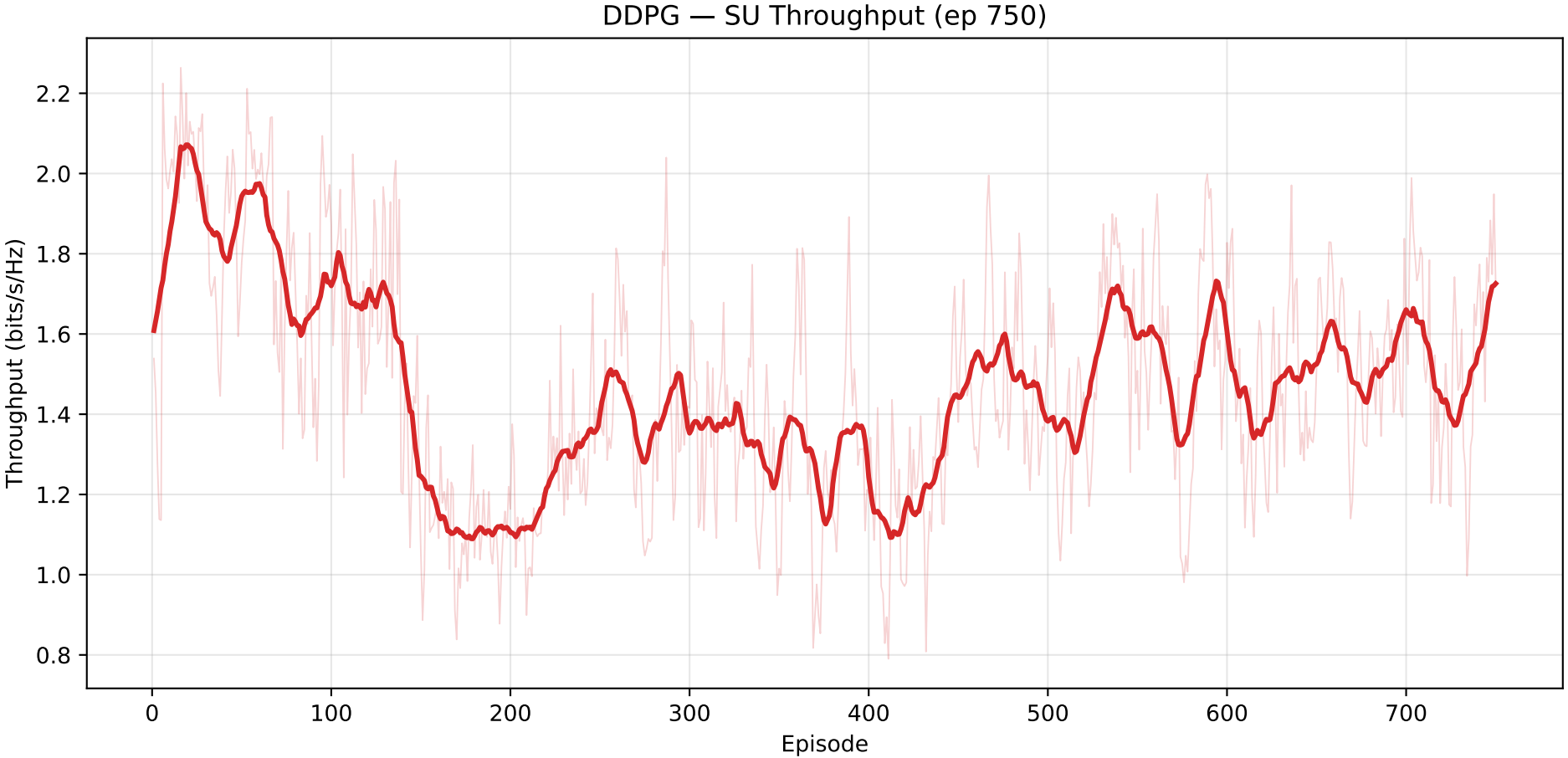
DDPG — Checkpoint Report (Episode 750)

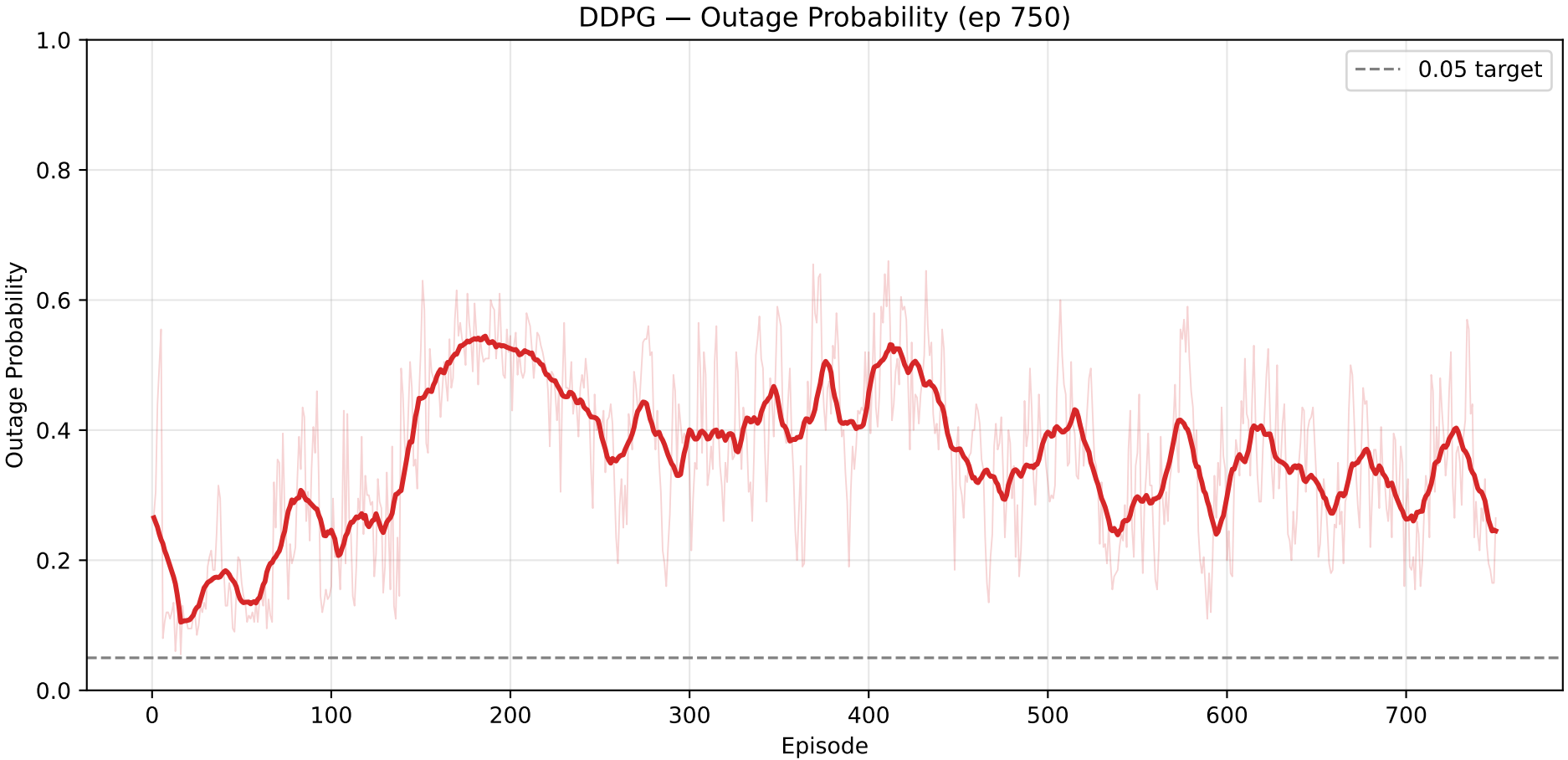
Nakagami-m=3.0 | P_MAX=3.0W | Steps/ep=750

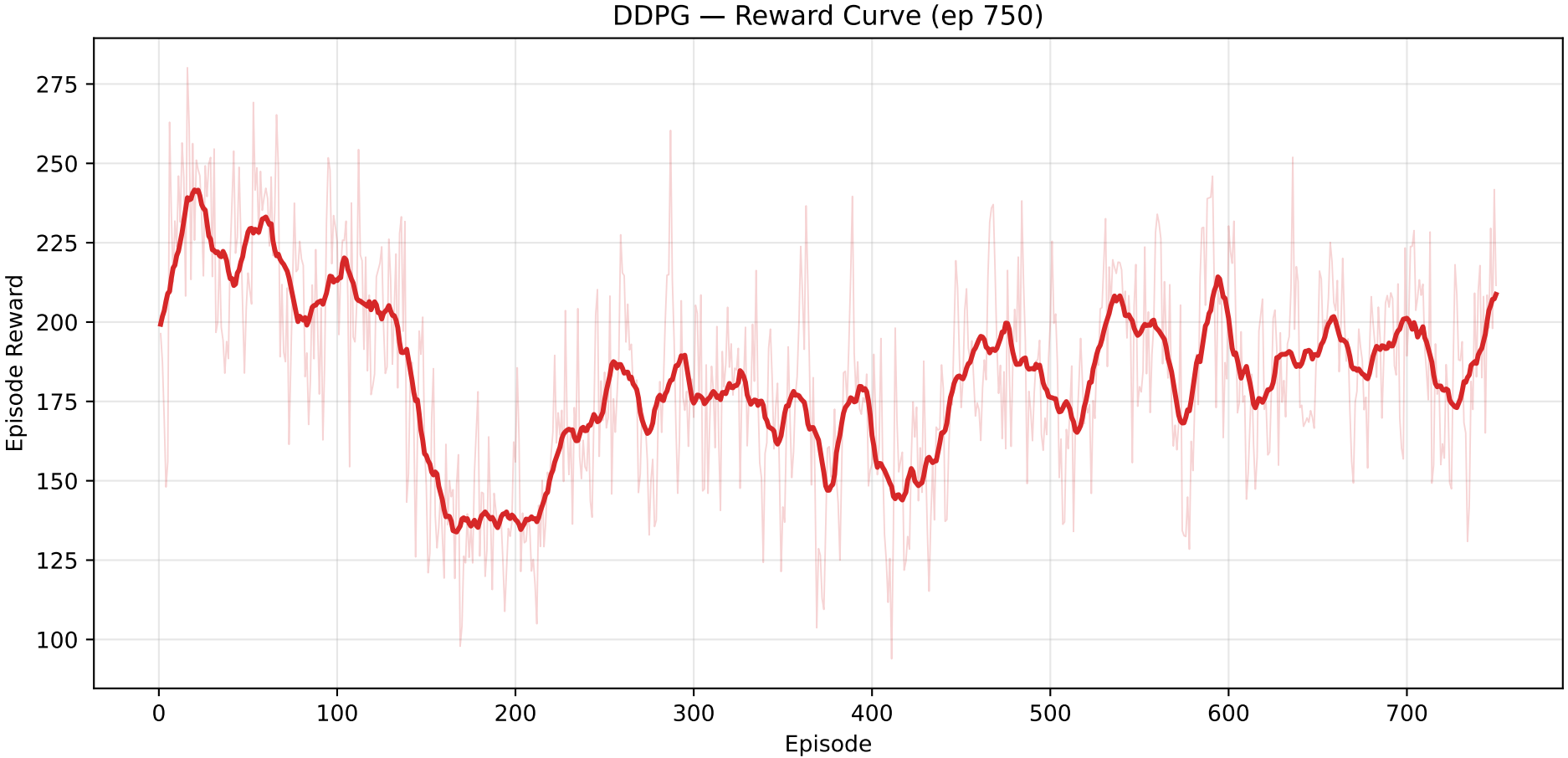
Metric	Value	Note
Episodes trained	750	
Avg Reward (last 100 ep)	191.4351	
SU Throughput (bits/s/Hz)	1.5481	
PU Throughput (bits/s/Hz)	1.0263	
Outage Probability	0.3150	5% target
Avg SU BER	0.072877	
Avg PU BER	0.144839	
Training Time (s)	0.0	


DDPG — SU SINR vs BER (ep 750)
BER
SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK (AWGN)
Nakagami-m=3 Theory
DDPG SU (simulated)
DDPG Mean BER


DDPG — PU SINR vs BER (ep 750)
BER
PU SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
Nakagami-m=3 Theory
DDPG PU (simulated)
DDPG PU Mean BER
PU SINR Threshold


DDPG — SU Throughput (ep 750)
Throughput (bits/s/Hz)
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0
100
200
300
400
500
600
700
Episode


DDPG — Outage Probability (ep 750)
0.05 target
Outage Probability
Episode
1.0
0.8
0.6
0.4
0.2
0.0
0
100
200
300
400
500
600
700


DDPG — Reward Curve (ep 750)
Episode Reward
Episode
275
250
225
200
175
150
125
100
0
100
200
300
400
500
600
700